

IoT Use Case : Singapore Smart City

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What is a Smart City

- A city that uses digital technologies, data, and intelligent systems to improve quality of life
- Integrates IoT, sensors, analytics, and automation into urban infrastructure

Key Characteristics

- Real-time monitoring of city services
- Data-driven decision-making
- Connected transportation, energy, buildings, and public services
- Citizen-centric, efficient, and sustainable operations

Why Smart Cities Matter

- Better mobility and reduced congestion
- Improved safety and public services
- Optimized resource usage and sustainability
- Enhanced living experience for residents



IoT Use Case : Singapore Smart City

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**SMART
NATION**
S I N G A P O R E



**Intelligent
Transport System
(ITS)**



**OneMotoring
portal**



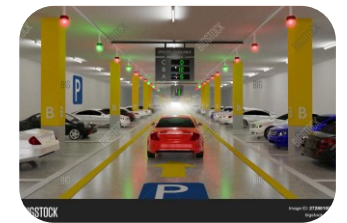
**Vehicle Recovery
Service (VRS)**



Your Speed Sign:



**Green Link
DEtermining
(GLIDE)**



**Parking Guidance
Service**



**Bus
Information
System**



**Smart Health
Assist**



**Smart Waste
Bins:**



**Smart light
sensors**



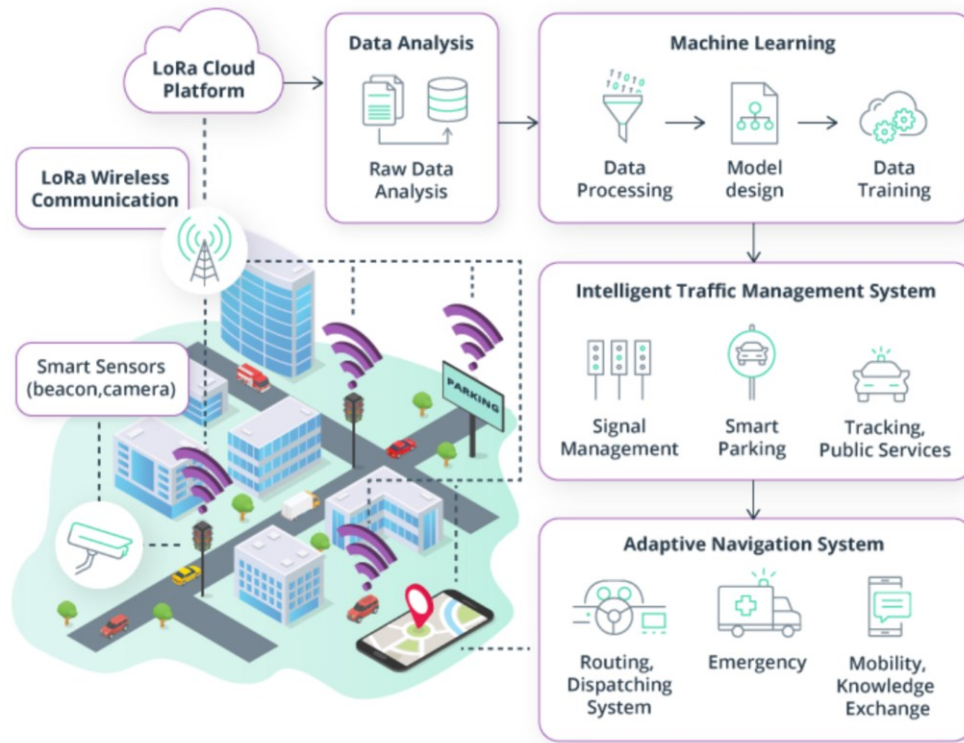
**Smart
electricity/w
ater meters**

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<https://itssingapore.org.sg/>

Singapore Smart City - Beneath the surface



Smart Sensors — The City's Eyes and Ears



LoRa Wireless Communication — The Low-Power Nervous System



LoRa Cloud Platform — The City's Data Highway



Data Analysis — Turning Raw Signals Into Insight



Machine Learning — The City's Predictive Brain



Intelligent Traffic Management System — Real-Time Urban Orchestration



Adaptive Navigation System — Guiding the City in Motion

SNSP Framework

5 interoperable layers

APPLICATIONS



INTEROPERABILITY



OPERATIONS & STANDARDS



DATA (PROCESSORS)



CONNECTIVITY



SENSING (DATA ACQUISITION)



Smart Nation Sensor Platform (SNSP)

What is SNSP?

- Nationwide, shared sensor and IoT infrastructure
- Integrates cameras, environmental sensors, and smart devices
- Centralized data platform for government agencies

Key Capabilities

- Real-time environmental monitoring (air quality, noise, weather)
- Public safety & situational awareness
- Smart lampposts with multi-sensor capabilities
- Unified data-sharing and analytics services

Why It Matters

- Enhances city operations & planning
- Enables faster incident detection and response
- Reduces duplication of sensor deployments
- Core pillar of Singapore's Smart Nation vision

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Core Components

- iTransport Platform: Unified backend integrating all ITS data for monitoring & incident management
- ITS Operations Control Centre: 24/7 monitoring of expressways & tunnels; coordinates incident response
- Smart Cameras & Sensors: Vehicle detection, classification, anomaly detection, and road-condition monitoring
- Smart Traffic Control: AI-enabled junctions & adaptive traffic lights for smoother journeys

Key Capabilities

- Real-time traffic information for public & motorists (EMAS signboards, MyTransport.SG, webcams)
- Faster incident detection & coordinated response
- Data-driven planning & policy development
- Supports Singapore's Smart Mobility 2030 vision

Why It Matters

- Reduces congestion & improves road safety
- Enhances commuter experience
- Enables efficient, sustainable urban mobility
- Considered one of the world's most advanced ITS deployments

Singapore's Intelligent Transportation System is nationwide network of sensors, cameras & traffic systems. It Collects real-time data on traffic flow, travel times & road demand to form a dynamic, real-time picture of mobility across the island



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What is Virtual Singapore?

- A dynamic, 3D digital twin of the entire nation
- Integrates real-time data, geospatial models, and simulation tools
- Built to support planning, research, and smart-city operations

Key Capabilities

- Simulate crowd movement, traffic flow, and emergency scenarios
- Analyze environmental factors (heat, wind, shading, noise)
- Visualize infrastructure, buildings, and urban layouts in detail
- Test future developments before real-world implementation

Why It Matters

- Enables data-driven urban planning
- Improves safety, resilience, and sustainability
- Supports government agencies, researchers, and city planners
- Core component of Singapore's Smart Nation vision

Virtual Singapore is one of the world's most advanced digital-twin platforms — a dynamic, 3D, data-rich model of the entire nation.

